

Will AI Image Synthesis Technology Help Constructivist Education at The Online Art Museum?

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ABSTRACT

Motivated by the outbreak of COVID-19, museums have increased interests in online museum and online museum education. This study presents a way to use AI image synthesis technology for online art education guided by a constructivist design approach. An experiment was conducted to empirically test the effectiveness of AI-based art education in the online museum context. A total of 83 participants visited one of 3 different web-based art museums (i.e., AI image synthesis not applied vs. AI image synthesis applied with given photos vs. AI image synthesis applied with self-uploaded photos). Those who experienced the online museum with synthesized images using self-uploaded photos reported a higher level of motivation and satisfaction and to experience in a more constructivist way compared to other conditions.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**; • **Applied computing** → **Interactive learning environments**.

KEYWORDS

Computer Vision, Education/Learning, Art, Quantitative Methods

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1 INTRODUCTION

Museums have been heavily impacted by COVID-19. According to a UNESCO report [35], 84% of museums were closed in 2020, 43% of museums were still closed in 2021, and access restrictions continue, resulting in an 80% decrease in museum revenue compared to 2019. During this challenging period, many museums have been focused on providing digital materials from previous exhibitions online, holding online exhibitions, and hosting workshops through social media [34]. According to the American Alliance of Museums (AAM) [26], 43% of museums enhanced access to digital collection resources.

A lot of art education materials, that used to be available offline, are now available online. The AAM report [26] shows that 75% of museums provide educational resources for children, parents and teachers and 60% of museums posted video lectures online. However, there seem to be important limitations in the available online art education materials. A lot of materials are available but the available content is not geared toward online education. Offline education materials are often designed for STEAM (Science, Technology, Engineering, Art, Mathematics) education [25, 26] and use art to enhance learners' expressivity, creativity, interest, and problem solving skills [27, 29]. These activities require active participation [1]. When practiced offline, the education programs can be guided by a teacher in an interactive fashion and tailored toward each student's need and interest [8]. However, when these materials are simply posted online, it is hard to expect that the users would participate in those activities as they would offline [19]. We believe mere digitization of education materials is not enough [7, 10, 18]; online education can provide interactivity and instant tailoring and can be further supported by AI technology.

Thus, this study proposes that AI image synthesis techniques, especially neural style transfer (NST) [9] with personalization, may

enable constructivist design in online museums. There are three goals for this study. First, to induce active participation through a constructivist design, we design a new web-based art museum and test its effectiveness. Secondly, we check the effects of personalization for constructivist design and give guidelines on how to design web-based art museums with NST. Lastly, provide a discussion of the potential of AI-based image synthesis technology in constructivist education.

2 RELATED WORK AND CONCEPTUAL BACKGROUND

2.1 Constructivism for museum education

Constructivism has been often identified as a solution to solve chronic communication issues in museums [13, 14, 16, 20]. Constructivist theory have been developed by many researchers since Piaget(1967) described personal constructivism [28] and it is still being developed as researchers are suggesting empirical and theoretical methodologies to encourage students and teachers to study in a learning-friendly environment. Its success may owe in part to educators' disappointment with behaviorist education methods [24]. Behaviorism-based education is authoritarian in its approach to students, placing the burden of learning solely on the shoulders of teachers [8, 24]. Thus, behaviorism-based education connects teachers and students vertically, so that they can carry out education by conveying completed knowledge to students.

On the other hand, the tenet of constructivism is that "knowledge is not passively received but built up by the cognizing subject" [36]. Thus, the learners in constructivism-based education aim to communicate with each other through experience-based opinions and form a horizontal communication that makes meaning. The double constructivist concept of dramatic consequences is described by Hein (1991). First, education should be centered on students, not on topics or lectures that need to be taught. Second, there is no independent knowledge of the meaning given from experience by the learner or the community of learners [12].

However, these properties assumed by constructivist education are rather difficult to implement in all teaching environments. First, from the educators' point of view, constructivist education takes a longer time and more human resources than traditional pedagogies. Second, educators must solve problems from a practical point of view so that learners can internalize more valuable knowledge by having the opportunity to examine themselves. In other words, they must be prepared to use not only one standard constructivist solution to problematic knowledge but also a variety of other approaches to facilitate learning [21]. Third, from the learners' point of view, intellectual dedication and perseverance are needed to learn with a constructivist education style. Furthermore, constructivist learning requires high cognitive ability on the part of learners [21, 23].

To successfully implement an online art museum, we believe that AI image synthesis technology has great potential to provide a possible solution to the aforementioned challenges of facilitating constructivist education. AI technology can adjust various contexts and tailor learning to each visitor to make it technologically possible to provide various learning experiences [4, 6]. The problem that only visitors with dedication and motivation can enjoy the

benefit of constructivist education can also be eased with constructivist design. Coming up with their own interpretation, thought, or even artwork based on their understanding is hard. However, AI image synthesis technology can offer enjoyable activities which are meaningful to each user and easily done by anyone. Web-based art museums equipped with AI image synthesis technology can induce active participation and museums do not have to re-organize programs depending on the visitor.

2.2 Personalization

Personalization is a method, that can be used in constructivist education to help learners play an active role in their learning by making links with their interests, passions, and aspirations [3]. When the activity is meaningful and relevant to learners, they will be more driven to learn and be involved in the learning process [3].

There are two types of personalization: static and dynamic. Static personalization requires a high level of user participation [30], in which users select content to use for personalization. Dynamic personalization collects user data and automatically provides personalized recommendations. This study conducted research based on static personalization.

Personalization can induce various effects. First, it connects learners' real-life with their education, helping them connect their education with their future [17]. Secondly, learning is effective when information is presented from the learner's point of view and related with their interests [2]. Lastly, personalization leads to a positive attitude toward content [15] "Affectively true self-as-source can foster positive attitudes toward content by invoking a greater sense of 'me-ness' in that users may see their content gate-keeping decisions as reflecting their identity" [32].

In this study, personalization is applied when using visitors' self-uploaded daily life photos for NST. This personalized process is expected to improve visitors' experience and induce a constructivist approach.

2.3 Design of CAAI

Considering the constructivist pedagogy and personalization effect, we designed CAAI (Create Art with AI) for web-based art museums.

Visitors upload at least 3 of their daily life photos to be used as content images, then choose one of Mucha's artworks to be used as style image, and finally create a new image which looks like the artwork's style has been applied. This application utilizes AI-based image synthesis technology, which is 'the neural style transfer' suggested by Gatys et al. [9]. Neural style transfer (NST) is a software algorithm which creates a new image using "content image" and "style image". We use visitor's photos as content images and artist's artworks as style images. Then NST creates an image that looks like the artist's style was applied, while the overall shape of the photo is maintained.

3 METHOD

3.1 Participants

The experiment was conducted to test the effectiveness of CAAI by conducting a controlled experiment to compare three conditions (Non-CAAI, CAAI with no personalization, and CAAI with personalization) using a between-subject design. In total, 83 university

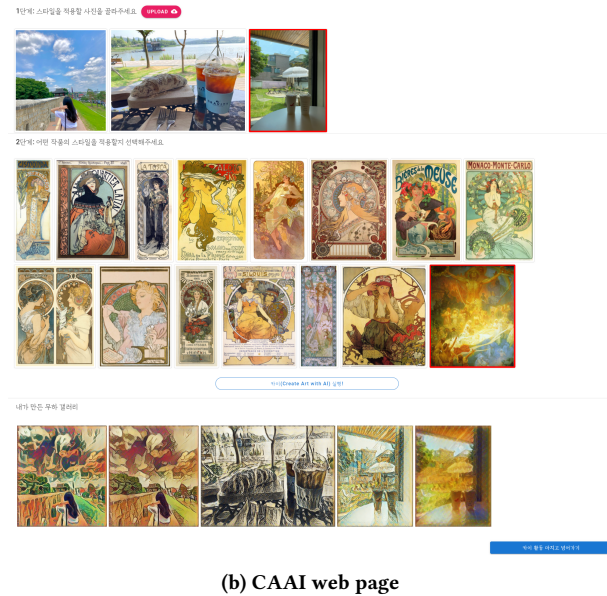
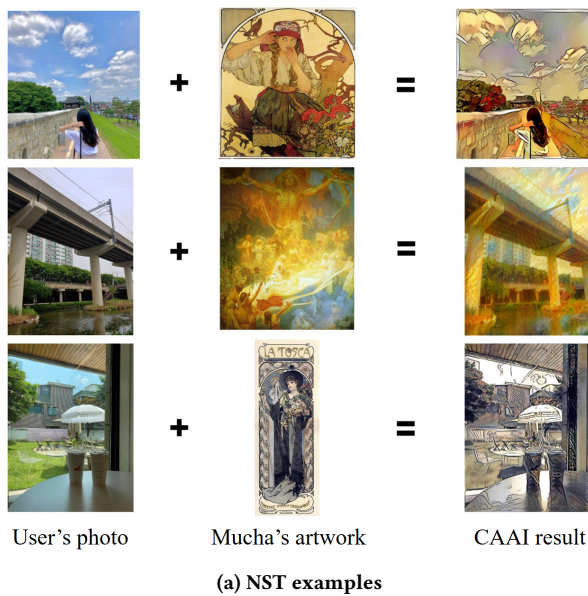


Figure 1: Image synthesis examples and user interface

students were recruited from the university website. All of the participants were randomly assigned to one of the three groups consisting of 30, 26, and 27 participants. Each participant received \$4 in exchange for their participation. This study was approved by the institutional review board (IRB) of the university.

3.2 Procedure

Before conducting the main experiment, a pre-test was conducted to decide which artist's artwork would be used in the main experiment. In total, 31 participants viewed seven artists' artwork to evaluate their awareness of each artist and their intention to learn more about artist. As a result of the pre-test, the more well-known an artist is, the greater the intention of people to learn. To avoid a possible ceiling effect, we chose the least well-known artist with the lowest score of intention to learn more about: Alphonse Mucha.

In the main experiment, participants received an email containing a URL link to the experiment material and instruction about the experiment. For the experiment, we created three different web-based art museums. The full version of the museum consists of three main steps. The first is to see Alphonse Mucha's artworks, the second is CAAI, and the last is to read the artwork descriptions. Depending on the assigned group, there is a difference only in the second step as shown in Fig 2. Our web-based art museum exhibits 15 artworks of Alphonse Mucha. Before the second step, participants were informed what kind of activity they were going to do. In the last step, the artwork descriptions included information such as the artist's intention, the type of artwork, and the historical context.

Participants accessed the website at any time they wanted. Those who were assigned to the Non-CAAI condition (control condition) only saw Mucha's artworks and descriptions without the other activity. This condition was intended to provide a similar experience to that which regular web-based art museums would provide.

CAAI with no personalization condition was conducted to test the effectiveness of personalization embedded in CAAI. Those who were assigned to this group were provided with three photos which were already picked by the researchers, as content image. We tested a few images in advance, and these three images were selected because the synthesized images came out with good results. The CAAI with personalization group was the experimental group. In this condition, participants experienced the activity in the program with photos they took. In order to see the personalization effect, participants in this group were asked to use their own memorable photos from daily life as a content image. We explained the example of an image that could work well as a content image before this activity, to help participants use CAAI smoothly.

Additionally, because it is a personal photo upload, we notified the participants that the photos would not be forwarded to our server and would only be temporarily stored within the user's browser. The user can upload photos from their device to the pop-up window. Then, the user can see at least 18 pictures at a glance, with photos and artworks shown in distinct regions. The user selects the photo and artwork one by one, and a red line appears to indicate the image that has been picked. When the user picks both the content image and the style image, the CAAI (Create Art with AI) button turns blue. When the CAAI button is clicked, a notification appears indicating that the process will take less than 10 seconds. Once the synthesis is complete, a pop-up window displays both the selected artwork and the synthesized photo. When the user closes the pop-up window, a synthesized image displays in the My Own Mucha Gallery area at the bottom of the screen, and users can choose to end or repeat the session after completing the activity once. If the user chooses to repeat, images continue to accumulate and be displayed in my own Mucha gallery.

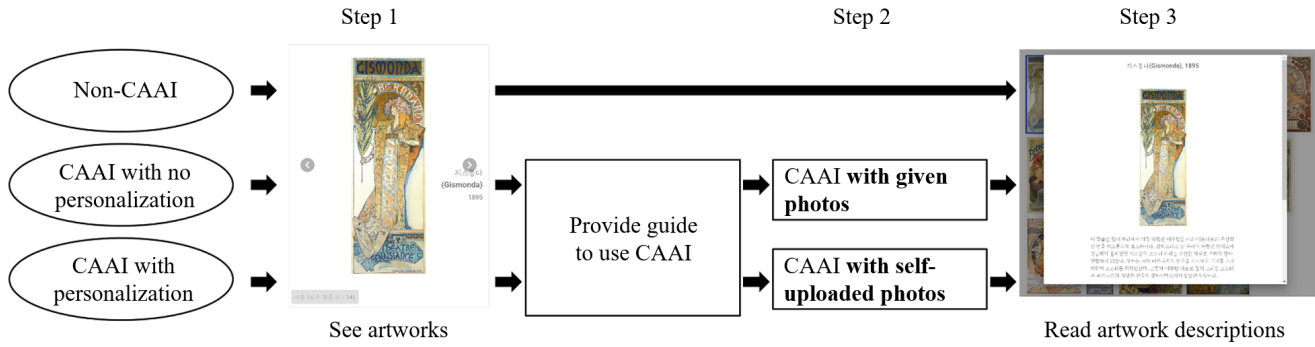


Figure 2: Differences in three conditions

3.3 Measures

The questionnaire included items measuring the following variables. Knowledge about art was measured with a 7-point Likert scale in answer to questions, such as “I have knowledge about art”, “I have a thorough education in art”, and “I am well aware of various painters” (Cronbach’s $\alpha = 0.88$). And the frequency of visits to art museums was measured with a year base (1 = don’t visit, 2 = once, 3 = twice, 4 = 3~4 times, 5 = 5~6 times, 6 = 7~11 times, 7 = more than 12 times).

Constructivist education variables were measured in this study using two main factors from [33] six and four items each: ‘Motivation toward reflections and concept investigation’ (Cronbach’s $\alpha = 0.89$) is about encouraging students to examine various perspectives and think reflectively, along with motivates students for further learning of related subjects. ‘Making meaning, real-life examples’ (Cronbach’s $\alpha = 0.77$) is about whether subject matter is relevant to students’ personal experiences, and the learning environment encourages students to think and make their own meaning. User experience was measured by the three items about revised version of perceived usefulness (Cronbach’s $\alpha = 0.91$) from [5], three items about revised version of perceived satisfaction (Cronbach’s $\alpha = 0.88$) from [22] and three items about revised version of intention to use (Cronbach’s $\alpha = 0.96$) from [11]. All of the responses were obtained on a 7-point Likert-type scale (1 = “not at all,” 7 = “very much”). Enjoyment was measured on a 9-point Likert-type scale with five items about revised version of enjoyment of the game (Cronbach’s $\alpha = 0.96$) from [31]. On the website, we collected data about how many times each participant conducted the CAAI activity and how many images each participant uploaded. Lastly, participants were asked a short-answer question about how they felt about this visit.

4 RESULTS

Before conducting the main analysis, demographic information was investigated. There were 58 females (70%) and 25 males (30%) and the mean age was 22.77 ($SD = 3.18$). Participants did not have much knowledge about art ($M = 2.93$, $SD = 1.48$) and their art museum visit frequency was also quite low ($M = 2.89$, $SD = 1.60$).

Table 1 shows the quantitative visitor experiences of each condition and whether there is significant difference between groups. The scores for the CAAI-P condition in terms of ‘Motivation toward reflections and concept investigation’ ($M = 4.91$, $SD = 1.23$) and ‘

Making meaning, real-life examples’ ($M = 5.24$, $SD = 1.27$) showed that participants felt CAAI-P was helpful in the two main factors of constructivist learning. The majority of participants (77.78%) of answered that they were somewhat satisfied, satisfied, or strongly satisfied with CAAI-P ($M = 5.59$, $SD = 1.45$). The score on ‘Enjoyment’ ($M = 6.92$, $SD = 1.48$) shows that the visit through CAAI-P was enjoyable, and the score for ‘Perceived usefulness’ ($M = 5.68$, $SD = 1.53$) shows participants think the visit through CAAI-P was beneficial and useful. Considering 5 or more points as agree (i.e., somewhat agree, agree, strongly agree), 14 participants (52%) in the CAAI-P condition responded intended to use CAAI-P even if they have to watch a 15-second ad in the middle of the activity.

To empirically test the effectiveness of CAAI-P in comparison to the other conditions, we conducted a one-way ANOVA test (Non-CAAI, CAAI with no personalization (CAAI-NP), CAAI with personalization (CAAI-P)) with post-hoc test.

First, the two main factors of constructivist learning were tested. The result of one-way ANOVA revealed that there were statistically significant differences among the three conditions in terms of ‘Motivation toward reflections and concept investigation’ ($F(2,80) = 3.27$, $p = 0.04$). Fisher LSD post-hoc test found that the mean value of ‘Motivation toward reflections and concept investigation’ was significantly different between CAAI-P and Non-CAAI ($p = 0.016$, 95% C.I. = [0.16, 1.54]). Similarly, three conditions were statistically significantly different in terms of ‘Making meaning, real-life examples’ ($F(2,80) = 5.88$, $p = 0.004$) and the Fisher LSD post-hoc test found significant difference between CAAI-P and Non-CAAI ($p = 0.0009$, 95%, C.I. = [0.46, 1.74]).

Second, user experience was tested. There was a statistically significant difference in ‘Perceived satisfaction’ ($F(2,80) = 6.73$, $p = 0.002$). The Fisher LSD post-hoc test revealed that the mean value of ‘Perceived satisfaction’ was significantly different between CAAI-P and Non-CAAI ($p = 0.001$, 95%, C.I. = [0.52, 1.97]), as well as CAAI-NP and Non-CAAI ($p = 0.007$, 95%, C.I. = [0.28, 1.74]). There was also a statistically significant difference in ‘Enjoyment’ ($F(2,80) = 4.47$, $p = 0.01$). The Fisher post-hoc LSD test revealed that the mean value of ‘Enjoyment’ was significantly different between CAAI-P and Non-CAAI ($p = 0.005$, 95%, C.I. = [0.43, 2.31]) and CAAI-NP and Non-CAAI ($p = 0.05$, 95%, C.I. = [0.01, 1.91]). Similarly, there was a statistically significant difference in ‘Perceived usefulness’ ($F(2,80) = 3.21$, $p = 0.046$). The Fisher LSD post-hoc test revealed that the

	Non-CAAI		CAAI with no personalization (CAAI-NP)		CAAI with personalization (CAAI-P)		Significantly different groups
	M	SD	M	SD	M	SD	
Motivation toward reflections	4.06	1.43	4.67	1.20	4.91	1.23	Non-CAAI vs. CAAI-P
Making meaning	4.14	1.08	4.74	1.28	5.24	1.27	Non-CAAI vs. CAAI-P
Perceived satisfaction	4.34	1.55	5.36	1.36	5.59	1.45	Non-CAAI vs. CAAI-P Non-CAAI vs. CAAI-NP
Enjoyment	5.55	1.90	6.51	1.93	6.92	1.48	Non-CAAI vs. CAAI-P Non-CAAI vs. CAAI-NP
Perceived usefulness	4.79	1.59	5.50	1.44	5.68	1.53	Non-CAAI vs. CAAI-P
Intention to use	4.21	1.87	5.33	1.69	5.09	1.53	Non-CAAI vs. CAAI-NP

Table 1: Mean, Standard deviation, and Significantly different groups

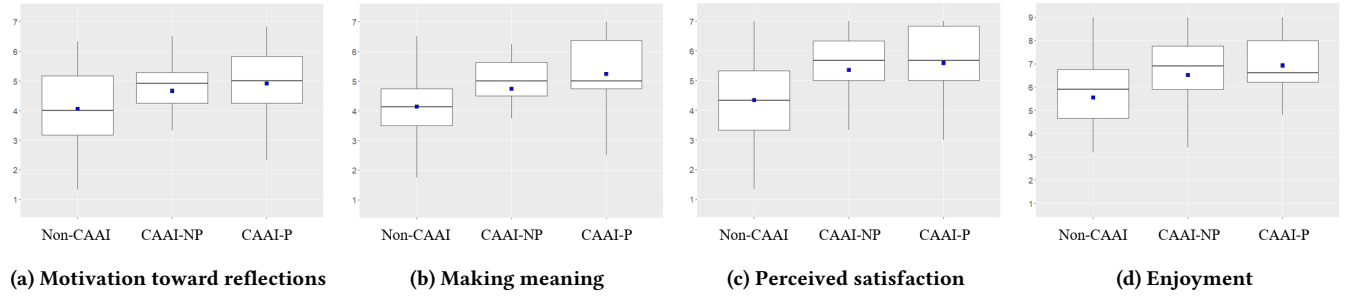


Figure 3: One-way ANOVA results (Blue dots are mean, and horizontal lines are median)

mean value of ‘Perceived usefulness’ was significantly different between CAAI-P and Non-CAAI ($p = 0.02$, 95%, $C.I. = [0.15, 1.63]$). There was a statistically significant difference in ‘Intention to use’ ($F(2,80) = 3.40$, $p = 0.04$). The Fisher LSD post-hoc test revealed that the mean value of ‘Intention to use’ was significantly different between CAAI-NP and Non-CAAI ($p = 0.02$, 95%, $C.I. = [0.21, 2.03]$).

Last, in the second step of the web-based art museum experiment, the CAAI-P group created images 6.11 times on average ($SD = 5.23$). Taking into account that participants could end CAAI after creating only one image, this indicates that participants enjoyed using CAAI-P. In addition, the CAAI-P group had to upload just 3 images to start creating images, but 10 participants (37%) voluntarily uploaded more images to use.

5 DISCUSSION

This study attempts to identify ways to allow online art museum visitors to take an active role to participate in reflective learning processes by applying constructivist design. Taking a constructivist design perspective, an AI-based image synthesis activity was applied in which users had opportunities to engage in a brief creation process by combining artwork displayed in the online museum and pictures of their own. AI vision technology was used to create synthesized artwork based on those two images. Through this process, users make active decision by choosing, linking, and imagining the possible results, while the actual creation process is taken care of by the system. It ensures active participation by providing multiple

opportunities to make choices but is quick and easy enough that even those with low competency in art creation can reap the benefit of the constructivist design. Further the effectiveness of personalization was tested with a controlled experiment to show that the system with personalization increased individual meaning-making for the visitors compared to the control group. In this section, we discuss implications of how CAAI-P affects visitors while looking around in online museums and provide design guidelines for future online museums.

5.1 Summary of the results

The results of the experiment showed positive user experience and learning effectiveness with CAAI with personalization (CAAI-P) and validated the effectiveness of the two sub-dimensions of constructivist education. ‘Motivation toward reflections and concept investigation’ is about encouraging student to examine various perspectives and think reflectively and to engage in further learning about related subjects. In this dimension, a significant difference was observed between the CAAI-P and control groups (Non-CAAI). Participants who were in the CAAI-P group were more motivated than those in the control group. The other sub-dimension of constructivist education was ‘Making meaning, real-life examples’, which relates to subject matter that is relevant to students’ personal experiences and in which the learning environment encourages students to think and make meaning. Again, a significant difference between the CAAI-P and control conditions was observed, with

those who were in the CAAI-P condition reporting a significantly more positive response in terms of making their own meanings and feeling that the program is about real-life. Results regarding user experience are also positive. The CAAI-P group reported being satisfied with and enjoying the online art museum. The results for the analysis of perceived usefulness showed a difference only between the CAAI-P and control groups, suggesting that using the personalization factor in CAAI plays an important role in making users feel the system is useful.

5.2 Design guidelines for future education programs in online art museums

This study provides several design guidelines. First, the findings showcase how the AI vision technology could be applied to online art museum and how the users learning experiences were like and how AI technology can encourage active participation which is critically important in constructivist education. AI technology can be utilized to reduce the burden on teachers, as it can easily and efficiently adjust multiple inputs to tailor toward each individual student. We believe that AI technology has a lot of potential in online education and the potential is not limited to art education. Second, this study showed that personalization is important and can be easily implemented via AI technology. Our results offer empirical evidence that personalization is effective to induce constructivist learning as well as increase enjoyment and satisfaction by creating a link between the subject and real-life. This finding has important implications for AI online education service designers. Using the novel AI technology can provide great learning outcomes. However, the results can be even more positive, when it is coupled with personalization. AI service designers should find a way to connect the education material with personally meaningful photos, pictures, music, or other artifacts. Last, although personalization is great, make sure to provide a test trial using the exemplar. In our experiment, the given images in the no personalization group were carefully prepared to pick the ones that provided the best results. Perhaps because of this reason, the qualitative results indicated that no one in the CAAI-NP group complained of difficulties in using the system. Furthermore, participants in the CAAI-NP group reported a higher level of intention to use compared to the CAAI-P group, despite the fact that we provided the same text-based explanations before using the CAAI. Therefore, we recommend providing reference content images first so users can learn by using those before they upload their own photos to use.

5.3 Limitations

The study has several limitations. Our sample was composed of university students and most of them were not well-versed in art appreciation. The results that we found in this study may not apply to individuals with different education levels or art interests. For example, actual online museum visitors may have a higher level of interest and related knowledge about art may show different experience patterns than shown in this study. Thus, it is necessary to investigate the effectiveness of the current version of CAAI-P with various groups. In addition, since the experiment was conducted with participants with low interest and knowledge of the artist and artworks, it is necessary to confirm through additional experiments

whether the results will hold even if the interest in the artist or artworks is not low. It is also necessary to investigate how much this constructivist online museum design affects online museum education through follow-up experiments.

6 CONCLUSION

With the prolonged COVID-19 pandemic, the importance of online museum education is increasing. This study investigated whether AI image synthesis can support constructivist education at the online museum environment. We presented CAAI which is well adapted to the context of online art museums, and showed that personalization through the process of using self-uploaded daily life photos to AI image synthesis improves user experience, motivates online museum visitors and gives them chances to construct their own meaning. We hope more studies will be conducted to further reveal effective education methods with various AI image synthesis technology.

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